Number Theory Tutorial

Fermat's Little Theorem, Wilson's Theorem and Primitive Roots

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Problem 1. Compute the following remainders using Fermat's Little Theorem:

(a) $3^{2026} \pmod{11}$;
(b) $7^{1234} \pmod{13}$;
(c) $2^{1000} + 5^{1000} \pmod{17}$.

Problem 2. Find the last two digits of 7^{2026}, that is, compute $7^{2026} \pmod{100}$.

Problem 3. Find all integers x with $1 \le x \le 30$ such that

$$x^{12} \equiv 1 \pmod{31}.$$

Problem 4. Use Wilson's Theorem to compute

$$16! \pmod{17}.$$

Then compute

$$1 \cdot 2 \cdot 3 \cdots 15 \pmod{17}$$

without directly multiplying all the factors.

Problem 5. Prove that if p is an odd prime, then

$$(p-1)! \equiv -1 \pmod{p}.$$

Use this result to determine the remainder of 10! modulo 11.

Problem 6. Determine the multiplicative order of each of the following elements:

$$\operatorname{ord}_{17}(3), \qquad \operatorname{ord}_{19}(4), \qquad \operatorname{ord}_{23}(5).$$

Justify that your answers are minimal.

Problem 7. Find all elements of order 6 modulo 19. How many such elements are there?

Problem 8. Determine whether 2 is a primitive root modulo 17. If it is, find all primitive roots modulo 17.

Problem 9. Find the smallest positive primitive root modulo 23. Verify your answer by computing the necessary powers.

Problem 10. Let p be a prime and let g be a primitive root modulo p. Determine all integers k with $1 \le k \le p-1$ for which g^k is also a primitive root modulo p.

Apply your result to list all primitive roots modulo 29.

Method for finding a primitive root modulo a prime p.

Factor

$$p-1 = q_1^{a_1} q_2^{a_2} \cdots q_k^{a_k},$$

where $q_1, \ldots, q_k$ are distinct primes.

An integer g is a primitive root modulo p if and only if

$$g^{(p-1)/q_i} \not\equiv 1 \pmod{p} \qquad \text{for every } i = 1, \ldots, k.$$

Once a primitive root g is found, all primitive roots modulo p are given by

$$g^k \pmod{p}, \qquad 1 \le k \le p-1, \quad \gcd(k, p-1) = 1.$$

In particular, there are $\varphi(p-1)$ primitive roots modulo p.